

Qualitative Cases: Semantic RAG Fails, TimeBound-RAG Recovers

Each row shows the question, gold answer, retrieved memories, and final model outputs.

Question
When is Alex's task after rescheduling?

Gold answer
2026-01-23 20:00

T1 · superseded **GOLD**
obs: 2026-01-15 09:00
event: 2026-01-21 18:00
Alex's task was scheduled for 2026-01-21 18:00.

T3 · expired
obs: 2026-01-16 02:00
event: 2026-01-17 02:00
Priya mentioned an unrelated task at the office.

T5 · expired
obs: 2026-01-17 18:00
event: 2026-01-23 18:00
Jordan mentioned an unrelated task at the library.

TimeBound-RAG retrieved

T4 · active **GOLD**
obs: 2026-01-17 09:00
event: 2026-01-23 20:00
Alex moved the task from 2026-01-21 18:00 to 2026-01-23

T2 · active
obs: 2026-01-16 01:00
event: 2026-01-16 01:00
Sam mentioned an unrelated meeting at the online room.

T1 · superseded **GOLD**
obs: 2026-01-15 09:00
event: 2026-01-21 18:00
Alex's task was scheduled for 2026-01-21 18:00.

X Semantic answer
2026-01-23 18:00

✓ TimeBound answer
2026-01-23 20:00

CASE 2: rescheduling

Question
When is Alex's delivery after rescheduling?

Gold answer
2026-01-17 20:00

Semantic RAG retrieved

T1 · superseded **GOLD**
obs: 2026-01-09 09:00
event: 2026-01-15 18:00
Alex's delivery was scheduled for 2026-01-15 18:00.

T7 · expired
obs: 2026-01-14 03:00
event: 2026-01-15 03:00
Alex mentioned an unrelated delivery at the online room.

T8 · expired
obs: 2026-01-14 20:00
event: 2026-01-19 20:00
Omar mentioned an unrelated delivery at the clinic.

TimeBound-RAG retrieved

T3 · active **GOLD**
obs: 2026-01-11 09:00
event: 2026-01-17 20:00
Alex moved the delivery from 2026-01-15 18:00 to 2026-01-17

T1 · superseded **GOLD**
obs: 2026-01-09 09:00
event: 2026-01-15 18:00
Alex's delivery was scheduled for 2026-01-15 18:00.

T26 · active
obs: 2026-02-11 21:00
event: 2026-02-13 21:00
Alex mentioned an unrelated task at the lab.

X Semantic answer
2026-01-15 18:00

✓ TimeBound answer
2026-01-17 20:00

CASE 3: rescheduling

Question
When is Omar's meeting after rescheduling?

Gold answer
2026-01-09 20:00

Semantic RAG retrieved

T1 · superseded **GOLD**
obs: 2026-01-01 09:00
event: 2026-01-07 18:00
Omar's meeting was scheduled for 2026-01-07 18:00.

T11 · expired
obs: 2026-01-15 17:00
event: 2026-01-21 17:00
Leo mentioned an unrelated meeting at the clinic.

T10 · expired
obs: 2026-01-14 03:00
event: 2026-01-15 03:00
Omar mentioned an unrelated review at the clinic.

TimeBound-RAG retrieved

T3 · active **GOLD**
obs: 2026-01-03 09:00
event: 2026-01-09 20:00
Omar moved the meeting from 2026-01-07 18:00 to 2026-01-09

T2 · active
obs: 2026-01-01 20:00
event: 2026-01-01 20:00
Priya mentioned an unrelated review at the airport.

T1 · superseded **GOLD**
obs: 2026-01-01 09:00
event: 2026-01-07 18:00
Omar's meeting was scheduled for 2026-01-07 18:00.

X Semantic answer
2026-01-21 17:00

✓ TimeBound answer
2026-01-09 20:00

CASE 4: rescheduling

Question
When is Nina's call after rescheduling?

Gold answer
2026-02-04 20:00

Semantic RAG retrieved

T1 · superseded **GOLD**
obs: 2026-01-27 09:00
event: 2026-02-02 18:00
Nina's call was scheduled for 2026-02-02 18:00.

T10 · expired
obs: 2026-02-12 01:00
event: 2026-02-14 01:00
Jordan mentioned an unrelated call at the client site.

T23 · expired
obs: 2026-03-03 21:00
event: 2026-03-08 21:00
Nina mentioned an unrelated appointment at the clinic.

TimeBound-RAG retrieved

T4 · active **GOLD**
obs: 2026-01-29 09:00
event: 2026-02-04 20:00
Nina moved the call from 2026-02-02 18:00 to 2026-02-04

T2 · scheduled
obs: 2026-01-28 00:00
event: 2026-02-01 00:00
Omar mentioned an unrelated report at the lab.

T3 · scheduled
obs: 2026-01-28 01:00
event: 2026-01-29 01:00
Priya mentioned an unrelated delivery at the lab.

X Semantic answer
2026-02-14 01:00

✓ TimeBound answer
2026-02-04 20:00